Supervised Admixture

2026-01-23

```
res_dir <- "../results/s05_b_supervised/"
```

Load gt

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt",
    reattach_to = "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld"
  )
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

Subset mad_bor_g1000_sum

```
moz_pop <- unique(grep("^Mozambique",
  mad_bor_g1000_sum$population,
  value = TRUE
))

pops <- c(
  "Borneo", moz_pop, "Betsimisaraka", "Merina", "Diego",
  "Sakalava_Tsimihety", "Betsileo", "Vezo", "Mikea", "Temoro"
)

mad_bor_g1000_sum_sub <- mad_bor_g1000_sum %>% filter(population %in% pops)

mad_bor_g1000_sum_sub <- mad_bor_g1000_sum_sub %>% group_by(population)
```

Write plink files

```
gt_as_plink(mad_bor_g1000_sum_sub,
  file = "../data/intermediate/admixture/admixture_supervised_subset",
  overwrite = TRUE
)
```

```
## [1] "../data/intermediate/admixture/admixture_supervised_subset.bed"
```

Write .pop file

```
mad_bor_g1000_sum_sub$Region <- "-"
mad_bor_g1000_sum_sub[mad_bor_g1000_sum_sub$population %in% c("Borneo", "Sumba", "Timor"), "Region"] <-
mad_bor_g1000_sum_sub[mad_bor_g1000_sum_sub$population %in% c("Betsimisaraka", "Merina", "Diego", "Saka
# whatever is left as "Other" set to Africa
mad_bor_g1000_sum_sub[mad_bor_g1000_sum_sub$population %in% moz_pop, "Region"] <- "African"

write.table(mad_bor_g1000_sum_sub$Region,
  file = "../data/intermediate/admixture/admixture_supervised_subset.pop",
  row.names = FALSE, col.names = FALSE, quote = FALSE
)
```

Run admixture

```
cd ~/Git/malagasy_diversity/code
bash s05_admixture_supervised.sh
```

```
## ****                  ADMIXTURE Version 1.3.0                 ****
## ****                  Copyright 2008-2015                     ****
## ****           David Alexander, Suyash Shringarpure,          ****
## ****                John  Novembre, Ken Lange                 ****
## ****                                                          ****
## ****                 Please cite our paper!                   ****
## ****   Information at www.genetics.ucla.edu/software/admixture  ****
##
## Random seed: 43
## Point estimation method: Block relaxation algorithm
## Convergence acceleration algorithm: QuasiNewton, 3 secant conditions
## Point estimation will terminate when objective function delta < 0.0001
## Estimation of standard errors disabled; will compute point estimates only.
## Supervised analysis mode.  Examining .pop file...
## Size of G: 299x198196
## Performing five EM steps to prime main algorithm
## 1 (EM)   Elapsed: 1.425  Loglikelihood: -6.45408e+07 (delta): 4.13285e+07
## 2 (EM)   Elapsed: 1.465  Loglikelihood: -6.30412e+07 (delta): 1.49958e+06
## 3 (EM)   Elapsed: 1.457  Loglikelihood: -6.26987e+07 (delta): 342510
```

```
## 4 (EM)   Elapsed: 1.49   Loglikelihood: -6.26021e+07 (delta): 96639.9
## 5 (EM)   Elapsed: 1.441  Loglikelihood: -6.25698e+07 (delta): 32275.3
## Initial loglikelihood: -6.25698e+07
## Starting main algorithm
## 1 (QN/Block)     Elapsed: 2.774  Loglikelihood: -6.19368e+07 (delta): 633033
## 2 (QN/Block)     Elapsed: 2.891  Loglikelihood: -6.18794e+07 (delta): 57335.7
## 3 (QN/Block)     Elapsed: 4.964  Loglikelihood: -6.18528e+07 (delta): 26657.8
## 4 (QN/Block)     Elapsed: 5.029  Loglikelihood: -6.18204e+07 (delta): 32339
## 5 (QN/Block)     Elapsed: 5.118  Loglikelihood: -6.17831e+07 (delta): 37353.5
## 6 (QN/Block)     Elapsed: 5.05   Loglikelihood: -6.17736e+07 (delta): 9515.09
## 7 (QN/Block)     Elapsed: 5.056  Loglikelihood: -6.17716e+07 (delta): 1987.27
## 8 (QN/Block)     Elapsed: 3.736  Loglikelihood: -6.17714e+07 (delta): 196.273
## 9 (QN/Block)     Elapsed: 3.786  Loglikelihood: -6.17714e+07 (delta): 22.0504
## 10 (QN/Block)    Elapsed: 3.867  Loglikelihood: -6.17713e+07 (delta): 8.40227
## 11 (QN/Block)    Elapsed: 3.794  Loglikelihood: -6.17713e+07 (delta): 0.38901
## 12 (QN/Block)    Elapsed: 3.75   Loglikelihood: -6.17713e+07 (delta): 0.0119854
## 13 (QN/Block)    Elapsed: 3.763  Loglikelihood: -6.17713e+07 (delta): 0.000283487
## 14 (QN/Block)    Elapsed: 5.324  Loglikelihood: -6.17713e+07 (delta): 2.27988e-06
## Summary:
## Converged in 14 iterations (75.838 sec)
## Loglikelihood: -61771344.461139
## Fst divergences between estimated populations:
##  Pop0
## Pop0
## Pop1 0.140
## Writing output files.
```

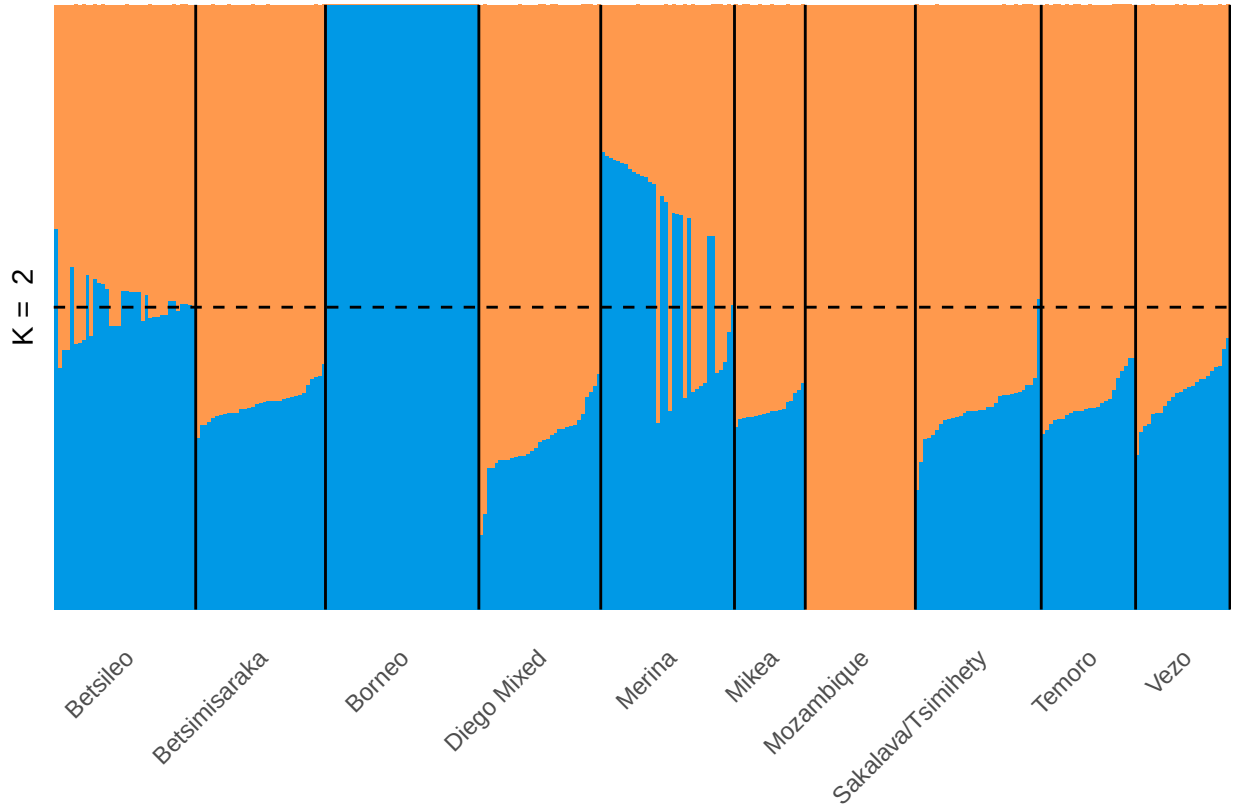
Read output

```r
predicted_q <- read_q_files("../data/intermediate/admixture")
```

Plot

```r
# rename populations
mad_bor_g1000_sum_sub$population <-
  ifelse(mad_bor_g1000_sum_sub$population %in% moz_pop,
    "Mozambique", mad_bor_g1000_sum_sub$population
  )
mad_bor_g1000_sum_sub$population[mad_bor_g1000_sum_sub$population == "Sakalava_Tsimihety"] <- "Sakalava
mad_bor_g1000_sum_sub$population[mad_bor_g1000_sum_sub$population == "Diego"] <- "Diego Mixed"

autoplot(get_q_matrix(predicted_q, k = 2, run = 1),
  data = mad_bor_g1000_sum_sub,
  annotate_group = TRUE, arrange_by_group = TRUE,
  arrange_by_indiv = TRUE, reorder_within_groups = TRUE
) +
  geom_hline(yintercept = 0.5, linetype = "dashed", color = "black")
```




```
ggsave(paste0(res_dir, "Admixture_supervised_malagasy.jpg"),
  width = 7, height = 6, dpi = 300
)
```

Recording R environment

Part of ensuring our workflow is reproducible is to keep track of the R environment. We will automatically save our R environment into a file called `r_environment.json`.